$U \subset \mathbb{R}^n$ open, bounded, ∂U smooth

Eigenvalue problem:

$$\begin{cases} -\Delta u = \lambda u, & x \in U, \\ u(x) = 0, & x \in \partial U \end{cases}$$

$A\vec{v} = \lambda \vec{v}$

where $\lambda \in \mathbb{R}$. In fact, $\lambda > 0$ $\left(-\int_U u \Delta u = \lambda \int_U u^2 \right.$

$$\left. \lambda = \frac{\int_U |Du|^2}{\int_U u^2} > 0. \right)$$

1D version:

$$-u''(x) = \lambda u(x) \quad 0 \le x \le L$$

$$u(x) = \sin\left(\frac{k\pi x}{L}\right), \quad k \in \mathbb{Z}, \quad \lambda = \left(\frac{k\pi}{L}\right)^2$$

Eigenvalues λ_k & eigenfunctions v_k depend on U.

Variational char. of eigenfunctions:

Let u be the first eigenfunction of $-\Delta$ on U. first = smallest λ

Claim: u minimizes the Rayleigh quotient

$$R[w] = \frac{\int_U |Dw|^2 \, dx}{\int_U w^2 \, dx} \quad \text{over } \mathcal{U} \setminus \{0\}$$

$$\mathcal{U} = \{ w \in C^2(U) \cap C(\bar{U}) : w = 0 \text{ on } \partial U \}$$

Take first variation:

$$R[u] \le R[w], \qquad w \in \mathcal{U} \setminus \{0\}$$

Choose $w = u + \tau\phi$, $\phi \in C^\infty$, compact support in U.

$$R[u] \le R[u+\tau\phi]$$

↳ Local min at $\tau = 0$:

$$0 = \frac{d}{d\tau} R[u+\tau\phi]\Big|_{\tau=0}$$

$$R[u+\tau\phi] = \frac{\int_U |Du + \tau D\phi|^2 \, dx}{\int_U (u+\tau\phi)^2 \, dx}$$

$$\frac{d}{d\tau} R[u+\tau\phi] = \Big[\int_U \cdots$$

$$\frac{d}{d\tau} R[u+\tau\phi] = \left[\int_U \frac{\partial}{\partial\tau}\left[|Du+\tau D\phi|^2\right] dx \int_U (u+\tau\phi)^2 dx - \int_U |Du+\tau D\phi|^2 dx \int_U \frac{\partial}{\partial\tau}(u+\tau\phi)^2 dx \right] \Big/ \left(\int_U (u+\tau\phi)^2 dx\right)^2$$

$$= \frac{\left[\int_U 2(Du+\tau D\phi)\cdot D\phi \, dx \int_U (u+\tau\phi)^2 dx - \int_U |Du+\tau D\phi|^2 dx \cdot \int_U 2(u+\tau\phi)\phi \, dx \right]}{\left(\int_U (u+\tau\phi)^2 dx\right)^2}$$

Plug in $\tau=0$:

$$0 = \frac{d}{d\tau}\Big|_{\tau=0} R[u+\tau\phi] = \frac{\left[\int_U 2\, Du\cdot D\phi \, dx \int_U u^2 dx - \int_U |Du|^2 dx \int_U 2u\phi \, dx \right]}{\left(\int_U u^2 dx\right)^2}$$

Simplify:

$$0 = \int_U Du \cdot D\phi \, dx \int_U u^2 dx - \int_U |Du|^2 dx \int_U u\phi \, dx$$

$$= \left(-\int_U \phi \Delta u \, dx + \cancel{\int_{\partial U} \phi Du \cdot \nu \, dS}^{\,0} \right) \int u^2 dx - \int_U |Du|^2 dx \int_U u\phi \, dx$$

Divide by $\int_U u^2 dx \neq 0$:

$$\boxed{\lambda = \frac{\int_U |Du|^2 dx}{\int_U u^2 dx}}$$

$$0 = -\int_U \phi \Delta u \, dx - \frac{\int_U |Du|^2 dx}{\int_U u^2 dx} \int_U u\phi \, dx = -\int_U \phi \Delta u \, dx - \lambda \int_U \phi u \, dx$$

$$= -\int_U \phi \underbrace{(\Delta u + \lambda u)}_{=0 \text{ in } U} dx = 0.$$

We've shown:

If u minimizes $R[w] = \dfrac{\int |Dw|^2 dx}{\int w^2 dx}$ over $\mathcal{U} \setminus \{0\}$,

then $-\Delta u = \lambda u$, where $\lambda = R[u]$.

Existence of a minimizer can be shown by the Direct Method of Calculus of Variations.

Higher Eigenvalues:

Recall: ① If $-\Delta u = \lambda u$, then $-\int_U u \Delta u = \lambda \int_U u^2 \Rightarrow \int_U |Du|^2 dx = \lambda \int_U u^2 dx$

$$\Rightarrow \lambda = \frac{\int_U |Du|^2 dx}{\int_U u^2 dx}$$

② If $\lambda_1 \neq \lambda_2$ and $\begin{array}{l} -\Delta u_1 = \lambda_1 u_1 \\ -\Delta u_2 = \lambda_2 u_2 \end{array} \Rightarrow \int_U u_1 u_2 dx = 0.$

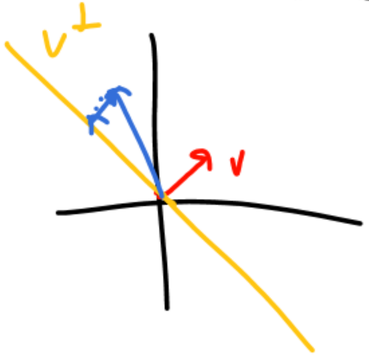
Let (λ_1, u_1) be the first eigenvalue of $-\Delta$ on U and corresponding eigenfunction.

Claim: The second eigenfunction u_2 minimizes $R[w]$ over $\left\{ w \in \mathcal{U} : \int_U w u_1 \, dx = 0 \right\}$

Notation: $\langle f, g \rangle = \int_U f g \, dx$.

To take first variation,

we don't have $R[u_2] \leq R[u_2 + \tau\phi]$ for general ϕ.



because $\langle u_1, u_2 - \tau\phi \rangle$ may not be zero.

Need to project $u_2 + \tau\phi$ onto

$$u_1^{\perp} = \{ w \in \mathcal{U} : \langle w, u_1 \rangle = 0 \}.$$

Let's assume $\int_U u_1^2 \, dx = 1$. (If not, divide by appropriate constant.)

For a smooth test function ϕ,

$$u_2 + \tau\phi - \langle u_2 + \tau\phi, u_1 \rangle u_1 \text{ is in } u_1^{\perp}. \text{ Check:}$$

$$\int_U u_1 \left(u_2 + \tau\phi - \langle u_2 + \tau\phi, u_1 \rangle u_1 \right) dx = \langle u_1, u_2 + \tau\phi \rangle - \langle u_1, u_2 + \tau\phi \rangle \int_U u_1^2 \, dx$$

$$= 0 \quad \checkmark$$

Then $R[u_2] \leq R[u_2 + \tau\phi - \langle u_2 + \tau\phi, u_1\rangle u_1]$

for all test functions ϕ, $\tau \in \mathbb{R}$.

$$\frac{d}{d\tau}\int_U (u_2+\tau\phi)u_1\,dx = \int_U \phi u_1\,dx = \langle \phi, u_1\rangle$$

Take $\frac{d}{d\tau} R[\overbrace{u_2+\tau\phi - \langle u_2+\tau\phi, u_1\rangle u_1}^{P_w}]$

$$= \frac{d}{d\tau}\frac{\int_U |Du_2 + \tau D\phi - \langle u_2+\tau\phi, u_1\rangle Du_1|^2\,dx}{\int_U (u_2+\tau\phi - \langle u_2+\tau\phi, u_1\rangle u_1)^2\,dx}$$

$$= \left[2\int (Du_2 + \tau D\phi - \langle u_2+\tau\phi, u_1\rangle Du_1)\cdot(D\phi - \langle \phi, u_1\rangle Du_1)\,dx \int_U (P_w)^2\,dx \right.$$

$$\left. - 2\int_U |D(P_w)|^2\,dx \int_U (u_2+\tau\phi - \langle u_2+\tau\phi, u_1\rangle u_1)(\phi - \langle \phi, u_1\rangle u_1)\,dx\right] \Big/ \left(\int_U (P_w)^2\,dx\right)^2$$

$$\frac{d}{d\tau}\Big|_{\tau=0} R[P_w] = 2\Big[\int_U \big(Du_2 - \cancelto{0}{\langle u_2, u_1\rangle} Du_1\big)\cdot\big(D\phi - \langle \phi, u_1\rangle Du_1\big)\,dx$$

$$\times \int_U \big(u_2 - \cancelto{0}{\langle u_2, u_1\rangle} u_1\big)^2 dx$$

$$- \int_U \big|Du_2 - \cancelto{0}{\langle u_2, u_1\rangle} Du_1\big|^2 dx$$

$$\times \int_U \big(u_2 - \cancelto{0}{\langle u_2, u_1\rangle} u_1\big)\big(\phi - \langle \phi, u_1\rangle u_1\big)\,dx\Big] \Big/ \Big(\int \big(u_2 - \cancelto{0}{\langle u_2, u_1\rangle} u_1\big)^2 dx\Big)^2$$

$$0 = \frac{2\int_U Du_2\cdot\big(D\phi - \langle \phi, u_1\rangle Du_1\big)\,dx \int_U u_2^2\,dx - \int_U |Du_2|^2 dx \int u_2 \underbrace{\big(\phi - \langle \phi, u_1\rangle u_1\big)}_{P\phi} dx}{\Big(\int_U u_2^2\,dx\Big)^2}$$

proj onto $u_1^\perp$

Simplify as above:

$$0 = \int Du_2 \cdot D\left(\phi - \langle \phi, u_1 \rangle u_1\right) dx - \underbrace{\boxed{\frac{\int_U |Du_2|^2 dx}{\int_U u_2^2 dx}}}_{=\lambda} \int_U u_2 \left(\phi - \langle \phi, u_1 \rangle u_1\right) dx$$

If $\langle \phi, u_1 \rangle = 0$, then

$$0 = \int_U Du_2 \cdot D\phi \, dx - \lambda \int_U u_2 \phi \, dx$$

$$= \int \phi \left(-\Delta u_2 - \lambda u_2\right) dx = 0$$

If ϕ parallel to u_1, $\phi = \alpha u_1$, $\alpha \in \mathbb{R}$,

$$\langle \phi, u_1 \rangle = \int_U \alpha u_1 \cdot u_1 \, dx$$

$$= \alpha \int_U u_1^2 dx = \alpha$$

$$\phi - \langle \phi, u_1 \rangle u_1 = \alpha u_1 - \alpha u_1 = 0$$

$$\int Du_2 \cdot D(\alpha u_1) = -\int u_2 \alpha \Delta u_1$$

$$= -\alpha \int u_2 u_1 \, dx = 0$$

Similarly,

$$\int_U \phi u_2 = \int_U \alpha u_1 u_2 \, dx = 0$$

$$\Rightarrow \int_U \left(D\phi \cdot Du_2 - \lambda \phi u_2 \right) dx = 0 \quad \text{in all cases}$$

$$(\phi \perp u_1 \text{ and } \phi \parallel u_1)$$

Integrate by parts as above: $-\Delta u_2 = \lambda_2 u_2$

$$\lambda_2 = R[u_2]$$

Similarly, λ_k is $\inf R[w]$ over $\{ w \in \mathcal{U} : \langle w, u_j \rangle = 0, \; j = 1, 2, \dots, k-1 \}$.

Smoothness of Eigenfunctions

$$-\Delta u = \lambda u$$

$$-\int u \Delta u = \lambda \int u^2$$

$$\int |Du|^2 dx = \lambda \int u^2$$

$$-\Delta(\partial_{x_j} u) = \lambda \partial_{x_j} u \implies \int |D\partial_{x_j} u|^2 dx = \lambda \int (\partial_{x_j} u)^2 dx$$

$$\leq \lambda^2 \int u^2 dx$$

$\implies$ Use Sobolev embedding.